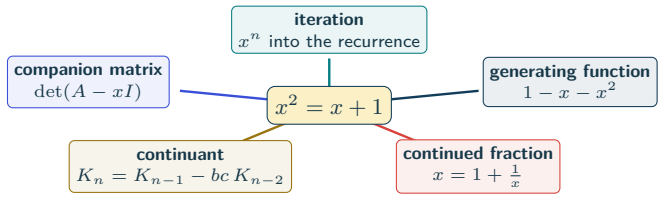
THE CHARACTERISTIC EQUATION ON ONE PAGE

The characteristic equation beneath five presentations

Five presentations of a recurrence are forced to one quadratic, and the diagonal gauge leaves it alone.

1 / THE RULE AND THE PICTURE

Five descriptions of the Fibonacci sequence are forced to the same quadratic



The recurrence $F_n = F_{n-1} + F_{n-2}$ has one characteristic equation, whose roots are $\varphi = 1.618$ and $\psi = -0.618$:

$$x^2 - x - 1 = 0$$

Every presentation carries these two roots: eigenvalues, Binet bases, convergent limit and pole reciprocals.

2 / THE FIVE PRESENTATIONS / HOW THE EQUATION APPEARS

Each presentation arrives at $x^2 - x - 1$ by its own route

presentation	route	what the roots become
iteration	try $F_n = x^n$, so $x^n = x^{n-1} + x^{n-2}$ and the equation appears	bases of the Binet formula
companion matrix	$A = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ has $\det(A - xI) = x^2 - x - 1$, the origin of the name	eigenvalues of A
continuant	$K_n = K_{n-1} - bc\,K_{n-2}$ with $bc = -1$ inherits the recurrence	growth rate of K_n
continued fraction	$[1; 1, 1, \dots]$ is the fixed point of $x \mapsto 1 + \frac{1}{x}$, so $x^2 = x + 1$	limit of the convergents
generating function	$\sum F_n x^n = \dfrac{x}{1 - x - x^2}$, the reversed polynomial	poles at $\frac{1}{\varphi}$ and $\frac{1}{\psi}$

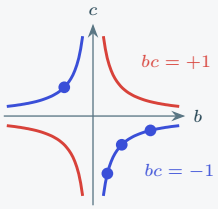
3 / HARD TO VARY / CHANGE THE EQUATION AND EVERYTHING MOVES

Altering a coefficient of $x^2 = px + q$ moves roots, bases, limits and poles together

equation	recurrence	first terms	dominant root
$x^2 = x + 1$	$a_n = a_{n-1} + a_{n-2}$	$1, 1, 2, 3, 5, 8$	$\varphi = 1.618$
$x^2 = 2x + 1$	$a_n = 2a_{n-1} + a_{n-2}$	$1, 2, 5, 12, 29, 70$ (Pell)	$1 + \sqrt{2} = 2.414$
$x^2 = x + 2$	$a_n = a_{n-1} + 2a_{n-2}$	$0, 1, 1, 3, 5, 11, 21$ (Jacobsthal)	2

4 / THE GAUGE FREEDOM / ONLY THE PRODUCT bc MATTERS

Diagonal conjugation slides (b, c) along a hyperbola $bc =$ const



For the continuant with 1 on the diagonal, b above and c below, $\det M_n = \det M_{n-1} - bc \det M_{n-2}$, so the pairs $(1, -1)$, $(2, -\frac{1}{2})$ and (i, i) all give Fibonacci because $bc = -1$. Conjugating by $D = \mathrm{diag}(d_i)$ sends $b \mapsto \frac{d_i}{d_{i+1}} b$ and $c \mapsto \frac{d_{i+1}}{d_i} c$, which slides the pair along its hyperbola. The only symmetric member is the complex pair (i, i), since a real symmetric or Hermitian continuant has $bc = b^2$ or $|b|^2 \geq 0$. Changing the product, $(i, -i)$ is Hermitian with $bc = +1$, giving $x^2 - x + 1 = 0$ and the period-six sequence $1, 1, 0, -1, -1, 0, \dots$

5 / THE GENERAL SEQUENCING LAW / PASCAL, STIRLING, WORPITZKY AND BELL

We meet a list, then methods, then one equation, and the triangles climb the same steps

Each row of a triangle is the previous row acted on by a bidiagonal operator, and the weights u_k and v_k are the only choices:

$$T(n, k) = u_k\,T(n-1, k) + v_k\,T(n-1, k-1)$$

triangle	(u_k, v_k)	row $n = 4$	sum	row sums
Pascal $\binom{n}{k}$	$(1,\ 1)$	1, 4, 6, 4, 1	16	powers of 2
Stirling $S(n, k)$	$(k,\ 1)$	1, 7, 6, 1	15	Bell numbers $1, 2, 5, 15, 52$
Worpitzky $W(n, k) = k!\,S(n, k)$	$(k,\ k)$	1, 14, 36, 24	75	Fubini numbers $1, 3, 13, 75, 541$

Scaling row k by $k!$, the diagonal gauge $D = \mathrm{diag}(k!)$, turns v_k into $k\,v_k$ and carries Stirling to Worpitzky while the diagonal $u_k = k$, which holds the eigenvalues $1, \dots, n$, stays put, the same invariance as the continuant. Worpitzky's numbers are the coefficients of x^n in Pascal's own basis, $x^n = \sum_k W(n, k)\binom{x}{k}$, and Pascal feeds Bell through $B_{n+1} = \sum_k \binom{n}{k} B_k$, so $B_4 = 1 + 3 + 6 + 5 = 15$.

ASIDE / THE NAME

The name comes from linear algebra, where $\det(A - \lambda I) = 0$ gives the eigenvalues of the companion matrix.

LINKS TO OTHER SHEETS

The gauge (i, i) and Pell, Jacobsthal: *Why i makes Fibonacci inevitable*; *Metallic means as differences of squares*. The continuant and eigenvalues: *The determinant in its many guises*; *Solving a quadratic as a line meeting a hyperbola*.

Convention. bc is the product of the off-diagonal entries, $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$. Compresses the essay *The Characteristic Equation*.

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